Linear Discriminant Analysis w/ $p=1$.

Fisher 1936.

Goal: Classify observations into one of K classes ($K \geq 2$), i.e., figure out

$$p_k(x) = \mathbb{P}[Y=k \mid X=x] \quad \text{posterior probability}$$

Environment:

- π_k ... prior probability that a randomly chosen observation falls into category $k = 1..K$
- $f_k(x)$ .... density function of X for observations from class $k = 1..K$

> $f_k(x)\,dx$ ... the probability that X falls into $(x, x+dx)$ for points in class $k=1..K$

Then,

Bayes Thm

$X \in (x, x+dx)$

$$\mathbb{P}[Y=k \mid X=x] = \frac{\mathbb{P}[Y=k] \cdot \mathbb{P}[X=x \mid Y=k]}{\boxed{\mathbb{P}[X=x]}}$$

The Law of total Probability